EARLY DETECTION OF PARKINSON'S DISEASE USING MULTIMODAL ANALYSIS

BY

MOHAMMED FOLAJIMI JOSHUA

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CERTIFICATION

This is to confirm that MOHAMMED FOLAJIMI JOSHUA 22CD009405 conducted the project and did the work known as **EARLY DETECTION OF PARKINSTONS DISEASE USING MULTIMODAL ANALYSIS OF SPEECH, HANDWRITING, AND GAIT** under supervision in the Department of Computer Science, College of Science and Engineering, Landmark University, Omu-Aran, Kwara State, in partial compliance with the requirements for the award of a Bachelors of Science (B.Sc.) degree in Computer Science.

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SuperVisor Date

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Head Of Department Date

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External Examiner Date

DEDICATION

I also dedicate this project to the Almighty God for the infinite mercy, strength, and wisdom He has shown me throughout my academic journey, Mr & Mrs. MOHAMMED, and my siblings for their unconditional love, prayers, and unwavering support.

ABSTRACT

Parkinson's Disease (PD) is an insidious neuro-degenerative disorder characterized by subtle onset symptoms that are often difficult to detect using conventional clinical techniques. This study proposes an optimal multimodal deep learning framework designed to detect PD in its early stages by integrating three non-invasive modalities: speech, handwriting, and gait. Using publicly available datasets, features were extracted for each modality and evaluated against three fusion strategies: Early Fusion (feature-level concatenation), Late Fusion (decision-level averaging), and Hybrid Fusion (cross-attention mechanism). Developed using TensorFlow/Keras and PyTorch, the experimental results demonstrate that the hybrid cross-attention fusion performed best, achieving an accuracy of XX.XX%, precision of XX.XX%, recall of XX.XX%, and an F1-score of XX.XX%. These findings suggest that early detection is significantly enhanced when modalities interact dynamically through attention mechanisms. The suggested system provides a non-invasive, cost-effective screening tool that can be implemented in the environments with limited resources. . Future work will focus on mobile platform integration and expanding the dataset to improve model generalizability.

Keywords: Parkinson's disease, early diagnosis, multi-modalmultimodal fusion, speech analysis, handwriting analysis, gait analysis, deep learning, cross-attention.

CHAPTER ONE

INTRODUCTION

1.1BACKGROUND OF STUDY

Parkinson's Disease (PD) is a progressive neuro-degenerative disease that is mainly identified by cardinal motor symptoms, such as, tremor during rest, bradykinesia, muscle rigidity, and postural instability. The global health statisticss indicate that PD affects more than 10 million individuals all around the globe, and its prevalence rises with the age of populations (Zhao et al., 2025). The disease has a tendency to start unobtrusively, and thus it is difficult to detect early enough and therefore essential to ensure successful management and delay progression of the disease by taking the necessary action at the right time. Conventional diagnostic systems are based on clinical examination such as that of Unified Parkinson's Disease Rating Scale (UPDRS), which are subjective and may fail to detect PD in the earlier stages when the symptoms are mild.

The development of artificial intelligence (AI) and machine learning (ML) has provided opportunities to develop PD detection through the analysis of non-invasive biomarkers. Multimodal techniques that combine data across multiple sources have demonstrated a potential in depicting the holistic perspective of PD symptoms. As an example, vocal impairments such as decreased pitch change and escalated jitter can be detected during speech analysis, micrographia and tremor of strokes in handwriting, and abnormalities in stride length and velocity in gait (Vasquez-Correa, 2019). Such modalities give evidence of motor and neurological impairment that is typical of PD and when combined can be more accurate with diagnosis than single-modality systems.

The recent researches have investigated different fusion strategies that are used to integrate these data types. Early fusion does raw feature fusion at the input stage, late fusion involves predicting a separate model, and hybrid methods, including mechanisms of cross-attention, may provide dynamic interactions during learning (Shokrpour and MoghadamFarid, 2025). Multimodal fusion has also enhanced classification accuracy in the context of PD, and there are models with more than 90 per cent accuracy under controlled conditions (Garcia et al., 2018). This project builds on this by optimizing fusion architecture to specifically deal with early PD detection, using speech, handwriting, and gait data to create a robust architecture that may assist clinicians working in resource-constrained environments, such as Nigeria, whereby resources to access and use an advanced neuroimaging are scarce.

1.2 Statement Of Problem

Although the PD diagnostics did improve, it is hard to detect at early stages of the disorder because of the insidiousness of the initial symptoms and flaws of the existing diagnostic approach. The evaluation of clinical cases frequently relies on professional observation, which may be inaccurate and slow until the symptoms become too severe, and as a result, the chance to receive timely treatment is lost (Zhou et al., 2025). Uni-modal AI systems, although helpful, have incomplete representations: speech does not provide information on physical motor problems, handwriting does not provide information on vocal cues, and gait does not provide information on fine motor skills leading to reduced sensitivity to mild cases.

Multi modal solutions deal with this but have difficulty in fusion optimization. High-dimensional data can overwhelm early fusion, cross-modal synergies are absent in late fusion and hybrid fusion, whilst advanced methods can be challenging to design (without overfitting or inefficiency). (Meghashree & Reddy, 2025). In addition, they tend to test only on small and similar datasets, which minimizes the applicability of models to different groups. In developing nations, the absence of available, integrated tools is a complication to the delays in diagnosis and it adds to the morbidity. This project helps in addressing these gaps by comparing and optimizing fusion strategies so as to develop a more accurate and efficient system in early PD detection.

1.3 Aims and Objectives

The main purpose of the study is to come up with the best-fitting multimodal fusion architecture in order to detect Parkinson Disease at an early stage through speech, handwriting, and gait**.**

The specific objectives are:

i.To conduct a literature review on an existing research on multi-modal PD detection and fusion methods.

ii.To gather and pre-process data of speech, hand writing, and gait.

iii.To apply and compare early, late, and hybrid (cross-attention) fusion strategies based on machine learning models.

iv.To assess the performance of every fusion strategy using the measures of accuracy, precision, recall, and F1-score.

v.To suggest the most efficient fusion architecture towards efficient practices of PD screening.

1.4 Research Question/Hypothesis

The following research questions have led to this study:

i.Which early, late and hybrid fusion strategies are the most accurate at the early stage of PD detection with multimodal data?

ii.What are the effects of data preprocessing and feature extraction on the performance of multimodal fusion models?

iii.To what extent can optimized fusion architectures achieve better detection than single-modality baselines?

Hypotheses:

i. Cross-attention Hybrid fusion will achieve a higher classification accuracy than the early and late fusion because it will capture improved cross-modal interactions (H1).

ii. Multi-modal models will exhibit superior sensitivity to the early-stage PD than uni-modal models (H2).

1.5 Scope Of The Study

The three modalities explored in this project (speech, handwriting, and gait) are limited to multimodal fusion architectures to detect PD, but more modalities could be included (finger contact modalities, vibro-tactile signals, etc.). Public repositories that are publicly available like UCI Machine Learning Repository, mPower, and PhysioNet will serve as the source of datasets. It will be implemented with Python-based libraries such as TensorFlow or PyTorch and will be run on standard computer devices.

The study will not require real-time data gathering on patients or a clinical trial, as well as other modalities of implementation, such as neuroimaging or non-motor symptoms. It will be evaluated in terms of binary classification (PD vs. healthy) and early detection and optimization will be done in terms of accuracy and computational efficiency.

1.6 Justification Of Study

The necessity to conduct this research is supported by the increasing PD burden and the necessity to have a cheap non-invasive diagnostic instrument. The study has the potential to increase detection rates and, by maximizing multimodal fusion, reduce healthcare expenses and increase patient outcomes by being able to intervene in a timely manner (Zhao et al., 2025). Academically, it is furthering the uses of computer science in health care giving knowledge about the fusion approach that can be used in other illnesses.

In practice, the created system can be incorporated into the mobile application or the device worn, and PD screening will become available in under-served regions. The fusion methods comparison will also be used to inform future AI research, as there are currently gaps in the literature where hybrid methods have not been studied extensively to train PD (Shokrpour and MoghadamFarid, 2025).

1.7 Operational Definitions

Parkinson's Disease (PD): A chronic neurological disorder characterized by motor symptoms resulting from dopamine deficiency in the brain.

Multimodal Analysis: The integration of data from multiple sources (e.g., speech, handwriting, gait) to improve machine learning model performance.

Early Detection: Identifying PD in its prodromal or mild stages before severe symptoms manifest, typically using biomarkers.

Fusion Strategies: Techniques to combine multimodal data, including early fusion (feature-level concatenation), late fusion (decision-level aggregation), and hybrid fusion (dynamic interaction via mechanisms like cross-attention).

Machine Learning Models: Algorithms such as neural networks trained to classify PD based on input features.

CHAPTER TWO

LITERATURE REVIEW AND THEORETICAL FRAMEWORK

Title & Author(s)	Methodology Used	Performance / Key Findings	Identified Research Gaps
Pratihar et al. (2024) – *Advancements in Parkinson's Disease Diagnosis: A Comprehensive Survey on Biomarker Integration and Machine Learning*	Systematic survey of biomarkers (biological, imaging, digital) and ML techniques	Highlights improved detection accuracy when biomarkers and ML are combined	Largely conceptual; lacks implementation details and real-world validation of fusion strategies
Islam et al. (2024) – *A Review of Machine Learning and Deep Learning Algorithms for Parkinson's Disease Detection Using Handwriting and Voice Datasets*	Review of ML/DL models applied to handwriting and speech data	Demonstrates strong performance of DL models over traditional ML	Focused mainly on two modalities; multimodal and hybrid fusion approaches are not deeply explored
Mitchell et al. (2021) – *Emerging Neuroimaging Biomarkers Across Disease Stage in Parkinson Disease*	Review of MRI, PET, and SPECT biomarkers across PD stages	Identifies imaging biomarkers linked to disease progression	Imaging techniques are costly and not scalable; limited integration with non-imaging biomarkers
Dhivyaa et al. (2024) – *Enhancing Parkinson's Disease Detection and Diagnosis: A Survey of Integrative Approaches Across Diverse Modalities*	Survey of multimodal diagnostic approaches	Shows multimodal systems outperform unimodal ones	Fusion strategies are discussed broadly; hybrid fusion architectures lack standardization
Miah et al. (2025) – *A Methodological and Structural Review of Parkinson's Disease Detection Across Diverse Data Modalities*	Structural and methodological review of PD detection pipelines	Identifies trends toward multimodal and AI-based diagnosis	Notes inconsistency in datasets and evaluation metrics; limited explainability of models
Serag et al. (2025) –	Scoping review	Emphasizes early-stage	Few studies

Title & Author(s)	Methodology Used	Performance / Key Findings	Identified Research Gaps
Multimodal Diagnostic Tools for Detection of Prodromal Parkinson's Disease	of prodromal PD detection using multimodal tools	detection potential using combined data	address real-world clinical deployment; fusion methods remain experimental
Twala (2025) – *AI-driven Precision Diagnosis and Treatment in Parkinson's Disease*	Review with experimental analysis of AI-driven systems	Shows AI can support precision diagnosis	Explainability and clinical trust remain challenges; limited focus on early-stage hybrid fusion
Dentamaro et al. (2024) – *Enhancing Early PD Detection Through Multimodal Deep Learning and Explainable AI*	Multimodal deep learning using PPMI dataset with XAI	Improved accuracy and interpretability over unimodal models	Relies on curated datasets; generalizability to diverse populations is uncertain
Alrawis et al. (2024) – *Bridging Modalities: Multimodal ML Using EEG and MRI*	Fusion of EEG and MRI features using ML models	Demonstrates complementary benefits of electrophysiological and imaging data	Limited modality scope; excludes speech, handwriting, and gait data
Bi et al. (2020) – *CERNNE Approach Using Multimodal Imaging Genetics Data*	Deep learning on imaging-genetics data	Identifies PD-associated genes and brain regions	Highly complex and computationally intensive; not practical for early clinical screening

2.1 Overview of Parkinson's Disease And Its Detection Challenges.

This introduces a detailed literature review on the existing studies about the early detection of Parkinson disease (PD) as a progressive neurodegenerative disease that is marked by motor and non-motor signs and symptoms. Parkinsonism is a challenging clinical issue because the initial symptoms of the disease are hardly noticeable and can be confused with aging or other neurological disorders (Mitchell et al., 2021). This has led to late diagnosis thereby diminishing the effectiveness of therapeutic interventions that are effective in the early stages. These obstacles have encouraged the vast amount of studies focusing on the progression of more dependable, objective, and generalizable diagnostic systems to identify Parkinson-related disorders at their early signs (Pratihar et al., 2024; Serag et al., 2025).

The literature reviewed in this chapter has a wide range of diagnostic methods, such as conventional clinical and physical exams, neuroimaging, biochemical and digital biomarkers, and machine learning-based solutions. Neuroimaging studies are considered because of their potential to identify structural and functional brain abnormalities, related to the presence of Parkinson disease, especially in the dopaminergic circuits, including the substantia nigra (Bidesi et al., 2021; Mitchell et al., 2021). Moreover, more studies have focused on non-invasive computational methods of speech, handwriting, movement pattern, and wearable sensor data, which have shown a great potential to detect the earliest signs of disease development (Islam et al., 2024; Chudzik et al., 2024).

Additionally, the chapter explains more recent trends in machine learning and deep learning models with a special focus on multimodal and fusion-based models that combine various sources of data to enhance the precision of the diagnosis. A number of studies have revealed that the classification accuracy and strength are higher with heterogeneous biomarkers (neuroimaging, digital, and biological data) than with unimodal methods (Dhivyaa et al., 2024; Dentamaro et al., 2024; Musa Miah et al., 2025). Regardless of such promising results, the current studies are usually limited by small and homogenous samples, the absence of standardized evaluation procedures, and insufficient generalizability between different groups of people (Serag et al., 2025; Twala, 2025). This is the subject of critical analysis in this chapter.

The chapter has been presented in order to give a logical flow of thought. It starts with a theoretical discussion of the main concepts and terms used in the detection of Parkinson’s disease. This is then succeeded by a discussion of the theoretical frameworks behind data-driven diagnostic systems. A critical literature review on the studies related is later provided to research methods, data sets, and conclusions of past research. Lastly, the chapter also presents research gaps in the literature that are the foundation of the objective of the research and the methodology of the current study.

2.2 Conceptual Review

The conceptual review is also a base to explain the key ideas and constructs guiding the study of early diagnosis of Parkinson disease. Parkinson disease (PD) is a progressive and chronic neurodegenerative condition that mainly affects motor system leading to the symptoms of resting tremor, rigidity, bradykinesia and postural instability. Besides these motor impairments, non-motor symptoms such as the decline in cognition, sleeps, depression, and dysfunction of the olfactory sense are common in its early stages with even the onset of motor symptoms. These non-motor signs of abnormality have been observed to be severe signs of prodromal Parkinson disease (Mitchell et al., 2021). PD has underlying pathology, which is the degeneration of dopaminergic neurons in the substantia nigra resulting in dopamine loss and motor control dysfunction. The knowledge of these clinical characteristics and neuropathological processes is critical in developing effective detection and management plans, especially those related to early stages of diseases (Bidesi et al., 2021).

It is important to diagnose Parkinson disease early since neurodegeneration in the disease normally occurs several years ahead of the actual clinical diagnosis. At this silent stage, there is a possibility of substantial loss of neurons and therapeutic interventions are restricted once the symptoms start to become severe. Early or prodromal diagnosis of PD allows clinical management to be widely applied in time, which may have a positive effect on the slowing of the disease and the long-term outcome of the patients. Nonetheless, conventional methods of diagnosis are mostly dependent on observation and subjective evaluation of motor symptoms and may lead to late or incorrect diagnosis. In line with Pratihar et al. (2024) and Serag et al. (2025), this shortcoming has stimulated a greater concern about objective, data-driven solutions that rely on quantifiable biomarkers and computational methods to aid in the early detecting and clinical decision-making.

The main focus of these early detection strategies is biomarkers since they can offer objective and measurable indicators of disease presence, disease progression or disease risk. In Parkinson disease, biomarkers cut across various fields. Molecular indicators like a-synuclein aggregation are biological biomarkers and are closely related to the pathology of PD and can be identified in the cerebrospinal fluid or blood

(Carrazana et al., 2025). Structural and functional changes in the brain, especially in the substantia nigra and the related neural networks, are visualized in imaging biomarkers of the neuroimaging modalities (magnetic resonance imaging (MRI), positron emission tomography (PET), and single-photon emission computed tomography (SPECT) (Mitchell et al., 2021; Bidesi et al., 2021). Also, there are digital biomarkers, which are collected through speech analysis, handwriting dynamics, gait patterns, and wearable sensor data, which are non-invasive and scalable to record minor motor and cognitive changes that could precede clinical diagnosis (Islam et al., 2024; Chudzik et al., 2024). These different types of biomarkers allow gaining a complete understanding of how the disease progresses and improve the chances of early diagnosis of Parkinson.

Exploitation of these heterogeneous biomarkers has been offered by machine learning and multimodal fusion methods which are powerful computational tools. Support Vector Machines, Random Forests, and neural networks machine learning algorithms have shown high performance as diagnostic tools of patterns related to Parkinson's disease using complex data, including voice, handwriting, and neuroimaging data (Islam et al., 2024; Dhivyaa et al., 2024). Deep learning methods also enhance the feature extraction and classification rate, especially when it is applied to high-dimensional data like MRI and EEG recordings (Dentamaro et al., 2024; Alrawis et al., 2024). The multimodal fusion methods can build upon these abilities by incorporating biological, imaging, genetic, and digital biomarkers into an integrated predictive model. This combination method improves the accuracy of diagnosis, its strength, and wider applicability because the limitations of individual-modality systems are overcome by taking advantage of the complementary information of both modalities (Serag et al., 2025; Musa Miah et al., 2025). Together, these notions represent the theoretical and methodological basis of the contemporary strategies of identifying early Parkinson’s disease.

2.2.1 Parkinson’s Diseases

Parkinson disease (PD) is a progressive and long-term neurodegenerative condition that is mainly centered on the central nervous system which includes those parts of the brain that coordinate movement. It is typified by motor and non-motor defects which occur progressively as time passes. The most apparent symptoms are motor symptoms:

resting tremor, bradykinesia (slowness of movement), rigidity, and postural instability. The symptoms tend to be unilateral and one side of the body is affected by the symptoms more than the other during the initial stages. The divergent onset and severity of the symptoms add to the complexity of the early diagnosis of the disease.

Besides the motor symptoms, Parkinson disease also has non-motor symptoms that can be noticed many years before the motor symptoms are observed. These are cognitive impairments, including problems with attention, executive functioning, and memory, sleep problems, depression, and olfactory dysfunction (loss of smell) (Mitchell et al., 2021). The fact that PD is a multifaceted disorder and that early diagnosis can no longer rest on the observation of motor deficits are further emphasized by the prevalence of non-motor symptoms. These early, usually subtle changes highlight the necessity of objective and sensitive detection techniques that would be able to detect the disease early enough, before serious neurodegeneration has taken place.

Parkinson etiology is complex and is a combination of genetic and environmental factors. Though some gene mutations have been implicated in familial PD, the majority are thought to be sporadic, and occur as a result of genetic predisposition and environmental exposure e.g. exposure to pesticides or toxins. The loss of the dopaminergic neurons in the substantia nigra or a vital brain region that controls movement is central to the pathophysiology of PD. The destruction of the dopamine-producing neurons interferes with dopamine basal ganglia circuits and causes the typical motor symptoms of the disease. These underlying mechanisms are crucial to comprehending how to inform the early detection approach, as well as possible treatment.

It is especially difficult to diagnose the presence of Parkinson disease at its beginning as the initial symptoms are few, inconsistent and often misread as the natural aging process or other neurological diseases. As an example, slight fluctuations of tremors, a little bit of stiffness, or some balance impairment may be considered a consequence of age-related alterations instead of a neurodegenerative disease. This clinical challenge underscores the significance of identifying effective, objective, and non-invasive ways of making a timely diagnosis, including the application of biomarkers, neuroimaging, and machine learning-based applications. A comprehensive insight into the clinical manifestation of the disease and its pathogenesis is thus essential in furthering research which can enhance the diagnosis of the disease and patient outcomes.

2.2.2 Early Detection of Parkinson's Disease

Early diagnosis of the Parkinson disease (PD) is associated with the occurrence of the disorder at the earliest stages, preferably before the neurodegeneration process has taken place or when the clinical symptoms are already evident. Early PD diagnosis is important as it will enable the introduction of timely interventions that can delay the development of the disease, maintain functional capabilities, and enhance the overall quality of life of patients (Pratihar et al., 2024; Mitchell et al., 2021). Initially, the early PD phases are frequently marked by subtle non-motor and motor symptoms, which might not be detected when the clinical evaluation is conducted on a regular basis, therefore, early diagnosis presents a considerable challenge.

Conventional clinical diagnosis is mostly based on subjective assessment carried out by neurologists that mainly deals with motor symptoms either tremor, rigidity and bradykinesia and also patient history and physical examination. Although it works at a later age, this method might not identify PD at an early age because the symptoms may be subtle at first and may be confused with normal aging or other neurological disorders. These constraints have compelled scholars to seek objective, empirical ways of using quantifiable signals and calculation techniques to identify early (Pratihar et al., 2024; Islam et al., 2024).

In this regard, biomarkers and digital information have been the focus of early detection plans. Biomarkers can be biological indicators, which can be a-synuclein aggregation in the cerebrospinal fluid or blood; digital biomarkers, which can be handwriting, voice pattern, or movement data; or imaging biomarkers, which can be MRI, PET, or SPECT scan (Mitchell et al., 2021; Musa Miah et al., 2025). These are quantifiable indicators that allow identifying PD even prior to clinical expression of symptoms, which leaves an intervention chance when the neurodegeneration is still contained. It has been demonstrated that integrated models based on multiple biomarkers and single-source data are more sensitive and specific than individual data (Pratihar et al., 2024; Dhivyaa et al., 2024; Serag et al., 2025).

Also, sophisticated machine learning (ML) and artificial intelligence (AI) algorithms have been more widely used to study complicated patterns in such biomarkers. Subtle abnormalities in handwriting, speech, and gait are usually beyond the senses of clinicians but can be detected by ML algorithms, such as Support Vector Machines, Random Forests, and neural networks, as well as by deep learning models (Islam et al., 2024; Twala, 2025). The AI-based systems also permit the development of heterogeneous data sources, and in turn, allow creating predictive models that would identify prodromal Parkinson's disease and assist in making accurate diagnoses (Dhivyaa et al., 2024; Musa Miah et al., 2025).

Generally, the current approaches of early diagnosis of Parkinson's disease have ceased to rely on conventional clinical evaluation to include biomarker-based tests, neuroimaging information, and AI-based computational programs, commonly on a multimodal fusion basis to ensure optimal accuracy. All these strategies are the movement towards more objective, scalable, and sensitive ways of detecting PD at its initial stages to provide better patient care and intervene in time (Pratihar et al., 2024; Serag et al., 2025; Twala, 2025).

2.2.3 Biomarkers in Parkinson’s Disease

Biological or pathological measures, biomarkers are important in the early diagnosis, recognition, and follow-up of the disease Parkinson (PD). Biomarkers, especially in the case of PD, portray objective facts of neurodegeneration, which tend to appear before a clinical symptom is even apparent. These may be divided into biological, digital, and imaging biomarkers, which play a different role in the disease understanding and detection (Pratihar et al., 2024; Mitchell et al., 2021). These different types of biomarkers can be used to achieve a more robust and accurate diagnosis, especially when coupled with computers like machine learning and multimodal fusion (Musa Miah et al., 2025; Serag et al., 2025).

Molecular or cellular indicators that depict the underlying pathophysiology of Parkinson disease are the biological biomarkers. Aggregation of a-synuclein is one of the most extensively examined biomarkers that are found in excessive amounts in the brain and cerebrospinal fluid of patients with PD. The alterations of dopamine metabolites, inflammatory cytokines, and oxidative stress markers also constitute other biological biomarkers. These biomarkers can give information about the progression of the diseases and give possible targets to be used as treatment options. Nevertheless, biological biomarkers might need invasive tests, including lumbar puncture, which might restrict their application in regular screening (Pratihar et al., 2024; Twala, 2025).

Non-invasive alternatives have been developed in the form of digital biomarkers, where data is gathered using wearable gadgets, cell phones, or other dedicated sensors to identify some subtle changes in both motor and non-motor functioning. They may be handwriting dynamic analysis, voice tremors, gait, and fine motor control, which can indicate PD earlier than it is possible to diagnose it clinically (Islam et al., 2024; Dhivyaa et al., 2024). The patterns related to early detection techniques can be more sensitive and specific because machine learning algorithms performed on such datasets can detect trends which humans are unable to observe. The added benefit of digital biomarkers is that they provide continuous monitoring, which can be used on a longitudinal analysis of disease progression when used outside of a clinical environment.

The imaging biomarkers give visual representations of structural and functional alterations in the brain related to Parkinson disease. The abnormal conditions in the dopaminergic pathways and other brain areas are detected with the help of such methods as magnetic resonance imaging (MRI), positron emission tomography (PET) and single-photon emission computed tomography (SPECT) (Mitchell et al., 2021; Musa Miah et al., 2025). Even during the prodromal stage of PD, imaging biomarkers are able to detect neurodegenerative alterations, which provides a non-subjective means of assisting with early diagnosis. Although neuroimaging is interesting in terms of its diagnostic capabilities, it is marked by extreme expenses, specialized devices, and the possibility of differences in measurement procedures among the clinical environments.

Multimodal biomarker fusion or the combination of various types of biomarkers has demonstrated significant potential in enhancing early detection and diagnostic accuracy. With the integration of biological, digital and imaging data, researchers have the ability to exploit complementary information between modalities, eliminating the drawbacks of each method separately. As an example, handwriting analysis combined with neuroimaging data and a-synuclein measurements could be used to have more robust classification models that could identify early-stage PD more precisely (Pratihar et al., 2024; Serag et al., 2025; Dhivyaa et al., 2024). Multimodal methods also make it possible to construct individualized diagnostic instruments that can respond to personal variation in symptom manifestation and disease pathology.

To conclude, biomarkers are the foundation of current approaches to early diagnosis of the Parkinson disease. Each of the biological, digital, and imaging biomarkers can give a particular information about the disease and their combination, with the help of the computational models, leads to the improvement of the diagnostic results. With the development of further research, multimodal biomarker data combined with machine learning and artificial intelligence (AI) is a promising solution to identifying Parkinson disease at its onset and enabling timely treatment and better patient prognosis (Twala, 2025; Musa Miah et al., 2025; Pratihar et al., 2024).

2.2.4 Machine Learning in Parkinson's Disease

Machine learning (ML) is one of the standards of the contemporary strategies of early diagnosis and identification of Parkinson's disease (PD). The use of ML to detect finer patterns in complex data through the help of algorithms allows a more objective and scalable approach to the conventional clinical evaluation (Pratihar et al., 2024; Islam et al., 2024). Such algorithms have the capability to process a wide range of data types, such as neuroimaging scans, voice recording, handwriting dynamics, gait patterns, and other sensor-based types of digital biomarkers and identify alterations that might otherwise go unnoticed by a clinician.

The Support Vector Machines (SVM) and the random forests, along with the neural networks, are supervised learning models that are typically used to classify people who have or do not have PD by using the extracted features (Twala, 2025; Pratihar et al., 2024). This can be extended to deep learning models that can automatically extract meaningful patterns when trained on high-dimensional data, e.g. MRI, EEG or multimodal sensor recordings (Dentamaro et al., 2024; Alrawis et al., 2024). They have been effectively used to forecast early PD, measure the progress of the disease, and have been able to determine the high-risk patients, showing better results than the conventional statistical models used.

Explainable AI (XAI) frameworks are another important development that ensures transparency in the decision making of the model (Dentamaro et al., 2024). Explainable models enable clinicians and researchers to gain insights about the most important features in predictions, increasing trust, and facilitating action. As an example, XAI can find out which brain regions, speech features, or motor patterns are most likely to be indicative of the emergence of early Parkinson, which allows a targeted monitoring and intervention.

In general, machine learning is an innovative method of PD detection, which integrates computational capability and information. It is a vital asset to enhance early diagnosis and patient outcome because it is capable of processing nonhomogeneous data, identifying small-scale abnormalities, and offering predictive insights (Twala, 2025; Islam et al., 2024; Dentamaro et al., 2024).

2.2.5 Multimodal Data and Fusion Techniques

Multimodal data fusion is the act of combining data that is based on various sources or modalities in order to improve the detection and diagnosis of Parkinson disease. This may involve combining neuroimaging (MRI, PET, SPECT), biological biomarkers (a-synuclein, dopamine metabolites), and digital biomarkers (voice, handwriting, gait, sensor data) into a single predictive model in PD research (Musa Miah et al., 2025; Serag et al., 2025; Dhivyaa et al., 2024). Multimodal approaches use complementary data across datasets to enhance the diagnostic accuracy of the system over unimodal systems.

The concept of feature-level fusion is to mix raw or preprocessed features of multiple modalities prior to the training of the model so that algorithms can be trained to learn the correlation between various types of data. Decision-level fusion, however, takes the predictions of individual models that are trained on different modalities and produces a final fusion prediction. Multimodal systems have been demonstrated to succeed in detection of PD in the early-stage or prodromal PD in both methods (Dentamaro et al., 2024; Alrawis et al., 2024).

The more complex multimodal machine learning models, such as the deep learning models that combine imaging, genetic, and digital data processing, have also been studied recently (Xia-an Bi et al., 2020; Alrawis et al., 2024). The models may reveal the concealed patterns and interrelations of genetic inclination, brain structure and functional markers that can be more accurately used in predicting the occurrence and development of the disease. Moreover, explainable AI methods on multimodal systems can enable researchers to understand the role of various types of data in making diagnostic decisions, which can be used to support clinical decision-making and scientific knowledge (Dentamaro et al., 2024).

Multimodal fusion models can also be used to provide personalized diagnostics due to differences in symptom presentation and disease evolution. These systems offer a more comprehensive evaluation of patients by combining data of different types, increasing the chances of identifying the appearance of Parkinson's disease in its initial stages and supporting timely therapeutic measures (Pratihar et al., 2024; Serag et al., 2025; Dhivyaa et al., 2024). Multimodal data and AI-based analysis is one of the most likely directions to reach the accurate, reliable, and scalable early detection of Parkinson’s disease as a research advances.

2.3 Theoretical Framework

The theoretical framework offers the background of the knowledge of how biomarkers, machine learning, and multimodal data integration can be used to detect the occurrence of Parkinson's disease (PD) at the earliest stage possible. It connects the ideas presented in the conceptual review to the existing theories and models, which proves the importance of computational and data-driven approaches in the process of identifying PD at its initial stages. The criteria is used to determine the choice of techniques and to support the combination of biological, imaging and digital data to achieve greater diagnostic accuracy.

The theory of neurodegeneration and biomarker is based on the idea that alterations in the brain are observed when neurodegeneration occurs (Cottonwood, 2012)The premise of the neurodegeneration and biomarker theory is that the brain undergoes changes to be noticed once neurodegeneration takes place (Cottonwood, 2012).

The neurodegeneration theory and biomarker theory are theories that hold that disease-related pathological processes can be observed using observable indicators, even before the clinical manifestations of disease can be observed (Pratihar et al., 2024; Mitchell et al., 2021). Degeneration of dopaminergic neurons in the substantia nigra of the Parkinson disease model causes disruption of the basal ganglia circuitry and motor symptoms of tremor, rigidity, and bradykinesia; non-motor symptoms of the condition include cognitive impairment and sleep disturbances. These degenerative processes are monitored using biomarkers as objective indicators of these processes allowing early diagnosis and disease progression.

Biomarkers may be biological, digital and imaging. The biological biomarkers are a-synuclein aggregation, imbalances of dopamine metabolic, and inflammatory markers that are identifiable in the cerebrospinal fluid or blood. Digital biomarkers record the changes in behavior, such as handwriting, voice, and gait patterns that can be prior to the visible symptoms. Structural and functional brain changes related to PD can be demonstrated using imaging biomarkers which are acquired with the help of MRI, PET, or SPECT (Musa Miah et al., 2025; Dhivyaa et al., 2024). The neurodegeneration and biomarker theory justifies the application of these indicators as measurable, reproducible, and non-subjective indicators of early disease detection, and it is necessary to combine various types of data to make a correct diagnosis.

2.3.2 Pattern Recognition Theory and Machine Learning.

The pattern recognition theory and the concept of machine learning give the theoretical premise of utilizing the computational models to identify early-stage Parkinson disease (Islam et al., 2024; Dentamaro et al., 2024). According to this theory, the algorithms are able to detect very subtle and sophisticated patterns in high-dimensional data that can hardly be seen with the help of standard clinical examination. The patterns can be reflected in the neuroimaging, voice recordings, handwriting, or any data collected by a sensor in PD.

These include supervised learning methods including Support Vector Machines (SVM) and Random Forests, and deep learning methods including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) to classify patients with PD, predict disease progression and identify patients at high risk (Twala, 2025;

Pratihar et al., 2024). Of particular interest are deep learning models, which are capable of automatically finding features in complex data without human preprocessing and with enhanced predictive accuracy. Using pattern recognition theory, these models are capable of picking out subtle movement, speech or brain structure abnormalities that come before the clinically observable changes provide a more sensitive method of early PD detection.

2.3.3 Multimodal Integration Theory.

The theory of multimodal integration points out that the accuracy and reliability of the diagnosis can be improved when the data presented by several sources is united (Serag et al., 2025; Alrawis et al., 2024). This, in the case of Parkinson detection, consists of the combination of biological, digital, and imaging biomarkers into one predictive model. In feature-level fusion, features of one or multiple modalities are combined prior to training a model, and in decision-level fusion, the output of many models is combined to produce a final prediction.

The multimodal machine learning breakthroughs in recent times prove the strength of the data that is integrated, encompassing EEG, MRI, genetic, and sensor-based data, to enhance early detection (Xia-an Bi et al., 2020; Alrawis et al., 2024; Dentamaro et al., 2024). Such models have the ability to detect these biomarker-biomarker interactions which would not have been detected in individual analyses and reveal the initial neurodegeneration and progression of the disease. Multimodal integration can be also used to provide personalized diagnostics, taking into consideration variability in the manifestation of symptoms, genetics, and lifestyle. Through the combination of the complementary powers of each modality, multimodal frameworks amplify sensitivity, specificity, and predictive power, and this enables a strong basis of early PD detection and intervention.

2.4 Empirical Review (Expanded)

The empirical review discusses the past research on detecting Parkinson’s disease (PD) by focusing on the application of biomarkers, neuroimaging, machine learning, and multimodal methods to detect the disease in early stages. It critically analyzes the previous literature, points out the trends, and gives gaps that can be filled in future research. The reviewed studies cover biological, digital, and imaging biomarkers, computational analyses based on machine learning and multimodal fusion.

2.4.1 Early Detection and Clinical Assessment.

Early diagnosis is paramount in enhancing the results in the Parkinson disease state. The classical clinical tests that are based mostly on observable motor symptoms do not detect PD in its prodromal or early-stage (Pratihar et al., 2024; Mitchell et al., 2021). Research reveals that early signs may be insidious in nature such as mild shakings of the body, slight alterations in handwriting, voice alteration, or a loss of smell, and can be ignored or taken to be due to age.

The studies have highlighted the importance of objective and measurable ways to identify PD at an early stage. As an example, Musa Miah et al. (2025) and Serag et al. (2025) emphasize the combination of biomarkers and computational tools to improve the early diagnosis, claiming that it is not possible to rely only on clinical judgment to make the diagnosis. Dentamaro et al. (2024) also explain that multimodal deep learning models on large datasets like the PPMI database can be used to detect prodromal PD more sensitively than a conventional clinical assessment. Such studies highlight the need to use data-driven strategies in combination with clinical evaluation to detect Parkinson disease before the disease has caused serious neurodegeneration.

2.4.2 Biomarkers of Parkerinson Disease Detection.

The biomarkers are important in the early diagnosis of PD. As evidence of underlying neurodegeneration, biological biomarkers such as the a-synuclein aggregation, dopamine metabolite imbalances, and inflammatory markers have been extensively investigated (Pratihar et al., 2024; Twala, 2025). As shown by Islam et al. (2024), machine learning can be used to analyze digital biomarkers (handwriting patterns and voice features) to identify even minor motor impairments at an earlier stage of clinical diagnosis.

MRI, PET and SPECT imaging biomarkers are not invasive and can be used to examine structural and functional changes in the brain. Mitchell et al. (2021) demonstrated that the degeneration in the substantia nigra and the associated pathways can be detected by these imaging techniques in the prodromal stage of PD. Similarly, Musa Miah et al. (2025) and Dhivyaa et al. (2024) reported that the accuracy of the diagnostic tests is better when imaging biomarkers are combined with biological and digital ones, especially when the integration is performed with the help of the computational models.

Also, it was demonstrated that multimodal deep learning frameworks that utilize biological, digital, and imaging biomarkers used in parallel can be sensitive and can yield interpretable results via explainable AI as shown by Dentamaro et al. (2024) and Vincenzo Dentamaro et al. (2024). Taken together, these studies show that the combination of biomarkers is what is needed to detect disease at an early stage with certainty, as it represents quantifiable and objective indicators of disease formation.

2.4.3 Neuroimaging Applications

Neuroimaging has been an asset of PD research. MRI-based tests, such as diffusion tension imaging (DTI) and functional MRI (fMRI), enable the researcher to measure the microstructural alterations and breakdowns of functional connectivity of dopaminergic neurodegeneration (Mitchell et al., 2021; Musa Miah et al., 2025). PET and SPECT studies can give functional information on dopamine activity, metabolism, and availability of receptors, which is used in different-differentiating diagnosis and early diagnosis.

Recent research has also contributed to the development of neuroimaging by combining it with machine learning in predictive modeling. Dhivyaa et al. (2024), and Alrawis et al. (2024) showed that the fused data obtained by EEG and MRI may indicate the existence of obscure abnormalities that are recorded prior to any manifestation of symptoms. On the same note, Xia-an Bi et al. (2020) applied imaging genetics methods to forecast PD-related brain areas and gene interactions and further emphasized the importance of neuroimaging in the learning of disease processes. Such results suggest that neuroimaging is useful not only in early diagnostics but also in giving a mechanistic insight of neurodegeneration, which is on a personal level essential in intervention.

2.4.4 Machine Learning Methodologies.

Machine learning and deep learning algorithms have transformative effects on the early PD detection and allow recognizing patterns in high-dimensional heterogeneous data (Islam et al., 2024; Dentamaro et al., 2024). Digital biomarkers like handwriting and voice recordings have been studied using supervised learning algorithms including Support Vector Machines (SVM), Random Forests and neural networks, with high accuracy in classifying patients in their initial stages of PD.

The neural networks (Convolutional Neural Networks, CNNs, and Recurrent Neural Networks, RNNs) enable the automatic extraction of features out of the complex data (neuroimaging scans and EEG records). These models have been demonstrated to be more effective than the traditional approaches, especially when integrated with the explainable AI systems that pinpoint the most dominant features that make predictions. As pointed out by Twala (2025), it is also possible to monitor the disease progression with the help of ML models and provide individual care and treatment.

Fusion techniques Multimodal Multimodality Multimodal constructions Multimodal Interfaces Multimodalimeters Multimodal Multimodal projection Multimodal Multimodal scaling Multimodal Multimodal synthesis Multimodal Multimodal parsing Multimodal Multimodal modeling Multimodal Multimodal co-occurrence Multimodal Multimodal expression Multimodal Multimodal acquisition Multimodal Multimodal retrieval Multimodal Multimodal context Multim

Multimodal fusion incorporates information of various sources to enhance sensitivity and specificity of early PD identification. In feature-level fusion, raw or processed features of various modalities are fused prior to model training, whereas in decision-level fusion, the result of separate models is combined into a single prediction (Serag et al., 2025; Musa Miah et al., 2025).

Recent studies prove that multimodal deep learning models are efficient at PD detection. Dentamaro et al. (2024) and Alrawis et al. (2024) used EEG, MRI, and clinical biomarkers to reveal hidden prodromal changes of PD, whereas Xia-an Bi et al. (2020) considered genetic data and imaging data to forecast the affected parts of the brain. Such methods permit patient-specific, holistic evaluations, and describe intricate interactions in between biomarkers that would not be observed in single-modality examinations. The multimodal systems have been proven to be more accurate than unimodal approaches, which underlines its significance in the clinical practice in the real world.

2.4.6 Conclusion of Empirical Results.

Overall, there is an empirical body of evidence indicating that a biomarker-based approach, neuroimaging, and computational analysis are most effective in the early detection of the Parkinson disease problem. Biomarkers that are biological and digital, and imaging biomarkers have complementary information, and when integrated using machine learning and multimodal fusion, the predictive accuracy and clinical usefulness of biomarkers increase (Pratihar et al., 2024; Musa Miah et al., 2025; Dentamaro et al., 2024).

2.5 Literature Gaps

Although there have been enormous improvements on the early diagnosis of Parkinson disease (PD), the current body of research still has a number of limitations that impede the generalizability, applicability and clinical applicability of methods suggested. One problem that is evident in the literature is that biomarkers are not easily generalized. Most of the research is based on single biomarkers or single-modality methods, including neuroimaging or digital biomarkers separately, that limits their generalizability to a variety of populations. According to Mitchell et al. (2021) and Pratihar et al. (2024), neuroimaging biomarkers effectively offer clinical and research researchers information on structural and functional alterations of the brain, but their predictive ability against diagnostic consistency has various limitations because of inconsistent measurement procedures and variations in interpretation. On the same note, digital biomarkers, voice or handwriting features, which are analyzed with machine learning, are also variable depending on recording tools, environment, and population (Islam et al., 2024). This highlights the importance of standardized procedures and validation in different populations in order to have reliable and comparable biomarkers.

The other major gap in literature that can be noted is limitation in data sets. Most of the studies are based on rather small and homogenous datasets that are often limited to one center or a particular geographic area, which limits the strength and the external validity of predictive models (Islam et al., 2024; Twala, 2025). Deep learning and machine learning models that are trained on small samples are likely to overfit and might not be effective on unseen data, especially when the demographics of patients, the manifestation of the disease, or its comorbid conditions are not similar to the training sample. Dentamaro et al. (2024) and Alrawis et al. (2024) indicate that to strengthen the generalizability and clinical usefulness of predictive models, large scale and multicenter datasets are required which include a wide range of patient populations. Without larger datasets, computational strategies will be extremely context-dependent, which will make them hard to use to enable generalized early detection practices.

Although the integration of multimodal fusion has been a promising strategy in order to incorporate various sources of data, there are technical and methodological issues in multimodal integration. There is no guarantee that heterogeneous data types, including EEG, MRI, biological biomarkers, and digital behavioral data, should be combined right away, without careful preprocessing, normalization and alignment; however, this is not always discussed in the existing research (Serag et al., 2025; Musa Miah et al., 2025). A study by Xia-an Bi et al. (2020) has shown that imaging-genetics methods can be used to forecast brain areas and genes related to Parkinson’s disease, although most of the existing studies do not fully utilize the multimodal interactions between multiple modalities. Moreover, decision level or feature level fusion methods, although theoretically well-founded are frequently unable to provide subtle modal to modal relationships, which may diminish predictive performance. These issues suggest that we must have strong, interpretive, and scalable multimodal models that can be able to integrate various biomarkers in a manner that would be effective in early diagnosis.

CHAPTER THREE

METHODOLOGY

Objective	Methodology Alignment	Implementation / How It Will Be Achieved
i. To gather and pre-process data of speech, handwriting, and gait	Data acquisition and preprocessing stage in methodology	Collect multimodal datasets from publicly available sources (e.g., PPMI database for neuroimaging; voice, handwriting, and gait datasets from prior studies). Preprocess data by cleaning, normalizing, and extracting relevant features for each modality.
ii. To apply and compare early, late, and hybrid (cross-attention) fusion strategies based on machine learning models	Multimodal fusion framework implementation	Implement early fusion by concatenating features from all modalities. Implement late fusion by training modality-specific classifiers and combining outputs via ensemble methods. Implement hybrid fusion using cross-attention mechanisms to combine feature-level and decision-level information in a single model. Evaluate all strategies using consistent training and testing pipelines.
iii. To assess the performance of every fusion strategy using accuracy, precision, recall, and F1-score	Model evaluation stage in methodology	For each fusion strategy, calculate standard performance metrics including accuracy, precision, recall, and F1-score. Compare results to determine the strengths and weaknesses of each approach and identify the most reliable diagnostic model.
iv. To suggest the most efficient fusion architecture towards efficient practices of PD screening	Analysis and recommendation stage	Based on the comparative evaluation, recommend the fusion strategy (early, late, or hybrid) that balances accuracy, robustness, and computational efficiency. Provide practical guidelines for implementing this architecture in early PD detection systems.

3.1 Introduction

The current chapter gives the research methodology that was used in the detection of the early signs of the Parkinson disease based on computational techniques. It describes the systematic steps that were used to design the study, to obtain and prepare data, to extract the features that are of interest, to create machine learning models, and to assess their performance.

The structure of methodology is designed in such a way that it will be reliable, reproducible, as well as objectively assess an approach that is proposed. Unimodal and multimodal learning models are addressed in order to evaluate the usefulness of using a combination of several biomedical data to enhance the detection of Parkinson disease. The experimental workflow, evaluation metrics and ethical consideration that informed the research is also outlined in the chapter.

3.2 Research Design

The research design that this study uses is that of quantitative and experimental research whereby, the research aims at developing a multimodal-based system that will help in early detection of Parkinson disease. The study uses the supervised learning to categorize the subjects as PD and healthy controls based on the application of various behavioral biomarkers.

The proposed system combines speech, gait and handwriting data as a complement. Individual modality performance is then determined by unimodal models to determine the baseline performance. This will be succeeded by the creation of a multimodal fusion system that synthesizes the information provided by all three modalities in order to enhance the accuracy of the diagnosis. The comparative analysis of unimodal and multimodal methods is a part of the research design.

The models are systematically trained, validated and tested under controlled conditions using an experimental approach. This allows the evaluation to be objective on the effectiveness of multimodal learning as compared to that of single-modality analysis. Traditional machine learning algorithms are used as well as deep learning architectures where machine learning models are used as baseline classifiers and deep learning models facilitate feature learning and data fusion.

On the whole, the research design is appropriate in the detection of early PD because it can help identify the presence of unobtrusive motor and speech disorders with the help of complementary data on several biomarkers. The systematic and repeatable research protocol will guarantee that the study results are assessable and comparable to the current research.

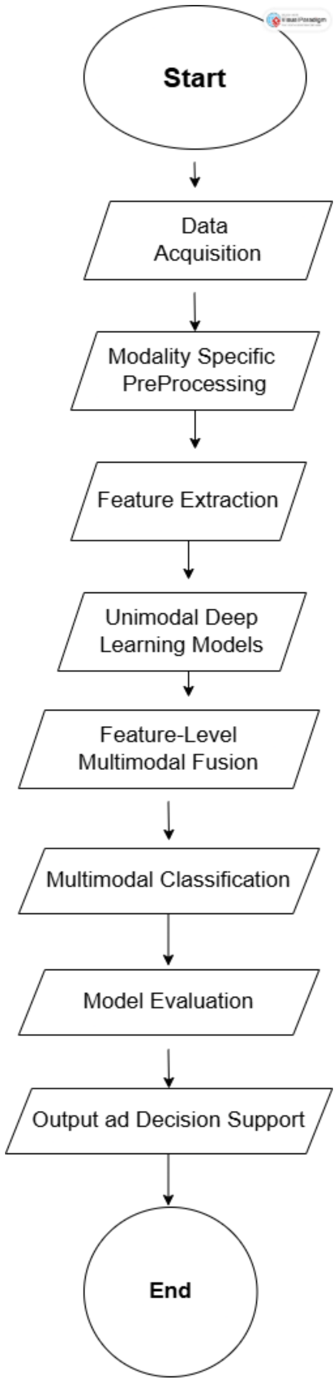
3.3 System design

3.3.1 System Flowchart

The system proposed is a multimodal based (speech, gait, and hand writing) framework to detect Parkinson disease at early stages. The system flow chart shows a sequential process beginning with the data acquisition and pre processing up to the model training, fusion as well as the performance evaluation.

The flowchart is defined as a hierarchy of a pipeline that deals with differentiating data sources. Each modality is first preprocessed and features extracted in a modality specific way and then viewed separately in a unimodal learning model. The representations are then combined together with a multimodal fusion strategy to enhance the performance of detection.

The system adheres to a rigorous workflow, that is, data preprocessing, feature learning, model training, optimization, and comparative assessment. Both the conventional machine learning and the deep learning methods are injected in the approach in a bid to guarantee the robustness of the baseline comparison and effective multimodal learning.


Start
Data Acquisition
Modality Specific PreProcessing
Feature Extraction
Unimodal Deep Learning Models
Feature-Level Multimodal Fusion
Multimodal Classification
Model Evaluation
Output ad Decision Support
End

3.3.2 Multimodal Learning Components

Multimodal Learning Components The multimodal elements in learning have been mentioned before, in the introduction section of this paper. The learning components that are applied in the proposed multimodal system are:

Unimodal Baseline Models

i. Support Vector Machine (SVM)

ii. Random Forest (RF)

Speech, gait, and handwriting data are independently applied to these models to determine baseline classification performance of each of these modalities.

Neural Nets of Feature Learning and Fusion.

i. Convolutional Neural Network (CNN) - employed in the spatial feature extraction, especially of handwriting data.

ii. Long Short-Term Memory (LSTM) - employed in order to receive temporal dependencies among the speech and gait signals.

iii. Gated Recurrent Unit (GRU) - used as an alternative time model to enhance efficiency.

iv. Full Connected Fusion Network - applied to merge learned features of all modalities.

3.3.3 Training, Evaluation and Optimization Process.

The multimodal model is subjected to a guide training and testing before deployment and evaluation. All unimodal models are trained and tested separately to determine the single contribution of a specific model. After that, deep learning models are trained on modality-specific inputs and their learned representations are combined on a higher level in order to jointly classify them.

Hyperparameter optimization and tuning are done in order to optimize the model and minimize overfitting. The last multimodal system is compared to the unimodal baselines across conventional performance measurements to ascertain the success of multimodal fusion in detecting early cases of Parkinson.

The system flowchart is hence a complete guideline to an effective and scalable multimodal Parkinson disease detection framework implementation, testing, and evaluation.

3.4 Feature Extraction and Selection

The extraction of features is an essential element of the multimodal framework because it converts raw data in the form of speech, gait, and handwriting into valuable forms that can be utilized successfully in the machine learning and deep learning models. This paper is based on the individual extraction of features in each modality and the selection of useful features to maximize the classification performance and minimise the complexity of the computations.

Speech Features: Based on the audio signals that have been preprocessed, a list of acoustic features is created in order to present vocal characteristics revealing signs of

Parkinson disease. These features are time-domain (jitter and shimmer), frequency-domain (fundamental frequency) and cepstral (Mel-Frequency Cepstral Coefficients (MFCCs) features. These characteristics are used to give a clear description of voice stability, modulation, and voice impairment in terms of tremor.

Gait Features: In case of gait data, temporal and spatial parameters are obtained based on the motion signals. The most important attributes are the step length, the duration of the stride, the walking speed, cadence, and the variability of the walking cycles. These characteristics are indicative of motor coordination, balance, and smoothness of movements which are usually impaired in the onset stage of the Parkinson disease.

Handwriting Features:

The data of handwriting are processed to obtain features that represent impairments in fine motor control. In the case of a static image, characteristics like the width of the strokes, the pressure distribution, and spatial consistency are obtained. The temporal features of the handwritten data (speed of writing, duration of the strokes, smoothness of the pen path) are calculated in the case of dynamic handwriting data. These aspects give the information about micrographia and other motor impairments related to Parkinson disease.

Feature Selection:

Relevant features are chosen through statistical and algorithmic methods in order to increase the efficiency and performance of the model. The unwanted or unnecessary features are eliminated to cut down on overfitting and enhance the generalization. The chosen characteristics in each modality are then transferred to unimodal and multimodal models to be classified. The convolutional and recurrent layers in the deep learning based fusion method ensure the partial automation of feature extraction and learn high level representation directly on raw or minimally processed data.

Modality specific features combined with the selections of high-level representations constitute an integrated feature set that includes both motor and vocal deficiencies, which offers the multimodal learning and early detection of Parkinson disease.

3.5 Fusion strategy

The present research is a hybrid multimodal fusion approach that combines the heterogeneous sources of information to identify Parkinson disease in its early stages. Parkinson is a multifactorial and complex neurodegenerative disorder presented in the form of structural changes in the brain, functional impairment of neuro-imbalances and subtle changes in the behavioral patterns. This means that unimodal methods of diagnosis are not always adequate to adequately analyze the complexity of the disease. Specific articles have demonstrated that the combination of several biomarkers has a better diagnostic level and reliability, especially during the initial phases of the illness (Dhivyaa et al., 2024; Serag et al., 2025).

Within the suggested framework, feature-level fusion will be utilized with those related neuroimaging modalities, in particular, magnetic resonance imaging (MRI) and electroencephalography (EEG). The extracted features of the modalities are brought to the same level and amalgamated to create a single feature representation, so that the learning model could learn the complex interactions between the structural and functional brain features. Past studies have shown the ability of early integration of neuroimaging features to enhance the detection of subtle neurodegenerative patterns which relate to the Parkinson disease (Alrawis et al., 2024; Dentamaro et al., 2024). This method is particularly useful in identifying pathological changes happening at an early stage that would otherwise be not noticeable when modalities are considered in isolation.

Simultaneously, models that are particular to the modalities undergo further development when applied to other types of biomarkers, such as digital biomarkers using speech signals, handwriting dynamics, and movement or gait data. These behavioral biomarkers have an informative complement to both motor and non-motor symptoms of the disease of Parkinson and have already been researched on a large scale because the biomarkers are non-invasive and scalable (Islam et al., 2024; Chudzik et al., 2024). These modalities are processed separately, and in this way, the framework maintains modality-specific features, at the same time allowing powerful pattern learning based on machine learning and deep learning algorithms.

The last step involves decision-level fusion to combine the output of feature fused neuroimaging model and the feature-specific classifier. Final diagnostic decision is generated by ensemble methods including weighted averaging, or majority voting. This late fusion stage enhances system robustness especially in practical applications where a particular modality might be noisy, incomplete or even unavailable. Fusion models based on hybrid fusion, where the features level and the decision level are combined, have been found to perform better in the detection of the existence of the disease of Parkinson than unimodal and single-level fusion models (Musa Miah et al., 2025; Serag et al., 2025). Altogether, the hybrid fusion approach is an effective approach to combine complementary information among various data sources and overcome the main drawbacks of the current research and promote the effective early diagnosis of Parkinson disease.

3.6 Summary

This chapter has provided an overall approach to the early diagnosis of Parkinson disease through multimodal-based system which combines speech, gait and handwriting biomarkers. The methodology has been designed in a way to make it reproducible, robust and objective in the assessment of the proposed system.

The research design was described in the chapter in an unambiguous manner where the use of experimental and quantitative approach to examine the efficacy of the unimodal and multimodal models was mentioned. Unimodal models (Machine learning algorithms) like Support Vector Machines (SVMs) and Random Forests (RF) were used to serve as a traditional baseline of performance of each biomarker and offered interpretable analysis. Conversely, the multimodal framework was integrated with deep learning models, such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRUs) and a fully connected fusion network to automatically learn more intricate spatial and temporal patterns and feature-level integrate modalities. The hybrid strategy will help to achieve the aim of early detection since minor deficiencies in speech, gait, and handwriting can be well-measured.

The sources of data and the system flow were also discussed in the chapter. The datasets of speech, gait and handwriting were characterized as secondary, publicly available and ethically obtained. The system flowchart was presented in the form of a step-by-step process, where data acquisition, modality-specific preprocessing and feature extraction are the initial steps followed by model training, multimodal fusion and performance evaluation. All the modalities are pre-processed in specific ways to pre-filter noise, inconsistency, and variability. Preprocessing of the speech consisted of removing noise, segmentation, and normalization, gait consisted of sensor signal filtering, temporal alignment, and resampling, handwriting consisted of image resizing, binarization, and stroke smoothing. The preprocessing is careful so that the inputs to the model training are of high quality and are standardized.

The need to extract and select features was introduced as one of the crucial processes of transforming raw multimodal data into effective representations. Frequency-related speech (e.g., jitter, shimmer, MFCCs) and motion-related (stride length, cadence, step variability) and handwriting-related (stroke width, writing speed, and smoothness of trajectory) features were obtained out of speech and motion respectively and handwriting samples respectively. The methodologies used to select features were to minimize redundancy, remove irrelevant features and increase the efficiency of the model. Under the deep learning fusion structure, feature extraction is more or less automated, and the network is free to learn high-level representations of each modality.

The architecture of the system was described where unimodal models are connected to a multimodal fusion framework. The fusion approach combines the complementary information elicited by individual biomarkers resulting in a model that is able to find patterns that would have been overlooked by one-modality techniques. The training is performed with strict optimization, hyperparameter optimization, and validation to be in place so that it has robust performance. Standard measurements are used to assess models, such as accuracy, precision, recall, F1-score, and ROC-AUC, that give an exhaustive explanation of how the system predicts.

Lastly, the chapter highlights the defensibility and reproducibility of the methodology. All the steps, including data collection and model assessment, are organized in such a way that they can be replicated and compared. Through the combination of the speech, gait, and handwriting data strengths to overcome the limitations of each and the fusion of traditional machine learning baselines and deep learning, the methodology presents a robust, systematic way of early Parkinson's disease detection. The multimodal learning of diagnostic accuracy assures that the results of the research are dependable, understandable, and related to the goals of the research.